

AI CA Assistant

Automated GST / ITR Summary Tool
for Indian Small Businesses

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Artificial Intelligence

FinTech

NLP / LLM

Problem Statement

Indian MSMEs lack affordable, accessible tools for GST and ITR compliance. Manual processes are error-prone, time-intensive, and inaccessible to businesses without dedicated CA support.

Introduction

An AI-powered web application that ingests business transaction data and automatically generates structured GST returns and ITR summaries, reducing dependency on manual accountancy.

Target Users

Small and micro enterprises, sole proprietors, freelancers, and first-generation business owners in India with limited financial literacy or CA access (Assumption: user base under GST registration threshold).

Expected Outcome

A functional prototype capable of parsing invoices/transaction CSVs, categorising tax heads, computing GST liability, and generating an ITR-ready summary report in PDF/Excel format.

Key Features

- Automated GST categorisation from invoice/CSV input
- ITR summary generation (Assumption: ITR-4 / ITR-3 scope)
- Natural language query interface for tax guidance
- Downloadable PDF / Excel compliance report
- Multi-lingual UI support (Hindi, English)

Competitor Landscape

- ClearTax – GST filing SaaS; no AI summarisation
- Zoho Books – Accounting suite; complex for micro-SMEs
- myBillBook – Billing focused; limited tax computation
- Differentiator: AI-first, conversational, zero CA dependency

Technology Stack

Frontend:	React.js / Next.js
Backend:	Python (FastAPI / Flask)
AI/NLP:	LLM API (assumption: OpenAI / open-source LLM)
Database:	PostgreSQL / Firebase
Reporting:	ReportLab / OpenPyXL

Future Scope

- Direct GSTN portal integration via official API
- AI audit-risk predictor and compliance score
- WhatsApp / chatbot interface for field users
- Support for TDS, advance tax, and MCA filings

COMPETITOR ANALYSIS

Dimension	AI CA Assistant ★ OUR PROJECT	ClearTax	Zoho Books
Industry Domain	Tax-Tech / FinTech AI-native, India-first	Tax-Tech / SaaS GST filing only	Accounting SaaS Mid-market focus
AI Usage	Core — LLM-driven NLQ + summarisation	Minimal Rule-based automation	Limited Predictive analytics
Social Impact	HIGH Targets unserved MSMEs	MEDIUM SME-focused product	LOW-MED Mid-market segment
Tech Complexity	HIGH NLP pipeline + GST logic	MEDIUM GST API integrations	HIGH Full ERP infrastructure
Innovation Potential	HIGH Conversational tax filing	LOW Mature product	LOW Established platform

Why we stand out: Only solution with an LLM-first conversational interface + zero CA dependency, purpose-built for Indian MSMEs. *Note: Competitor data from public sources; benchmarks require validation.*

DEVELOPMENT ROADMAP & EXPECTED OUTCOMES

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01 Research

Literature review, regulatory mapping (GST/ITR frameworks), tech feasibility

02 Data Collection

Anonymised invoice samples, GST return templates (Assumption: synthetic/mock data used for prototype)

03 Model Dev.

LLM prompt engineering, classification pipeline, GST logic layer implementation

04 Testing

Unit & integration tests, accuracy benchmarking on edge cases, user acceptance testing

05 Deployment

Cloud hosting (assumption: AWS/GCP free-tier), CI/CD pipeline, staging environment

06 Evaluation

Performance metrics: accuracy, latency, user satisfaction (SUS score); iteration cycle

Expected Benefits: Accurate GST/ITR summary output | Reduced compliance time by ~60% (projected) | Accessible to non-accountant users | Exportable reports for CA review